

Focal archaic-carrier coalescence

$s = 0.002$ | saved simulations, newly decoded with Gamma-SMC

2% introgression pulse + selection starting at 50 kya

100 selected regions + 1,000 neutral regions

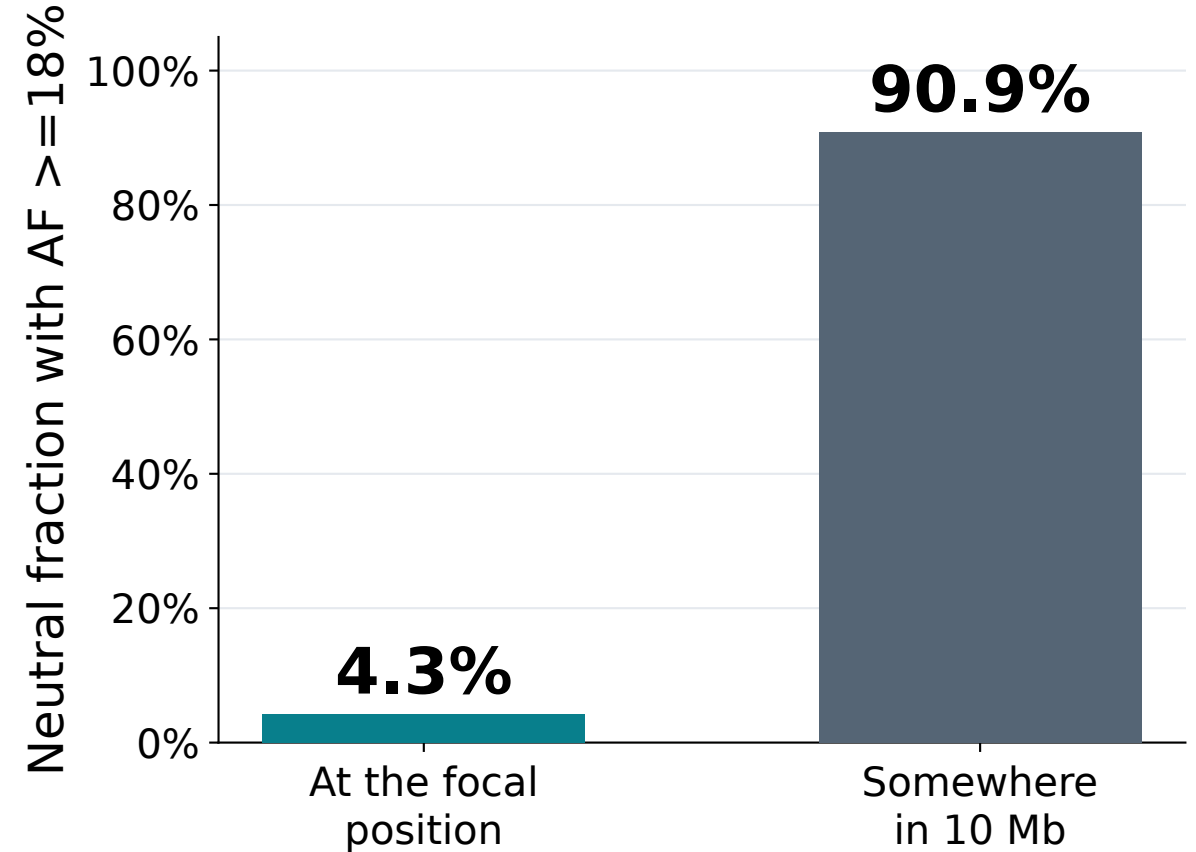
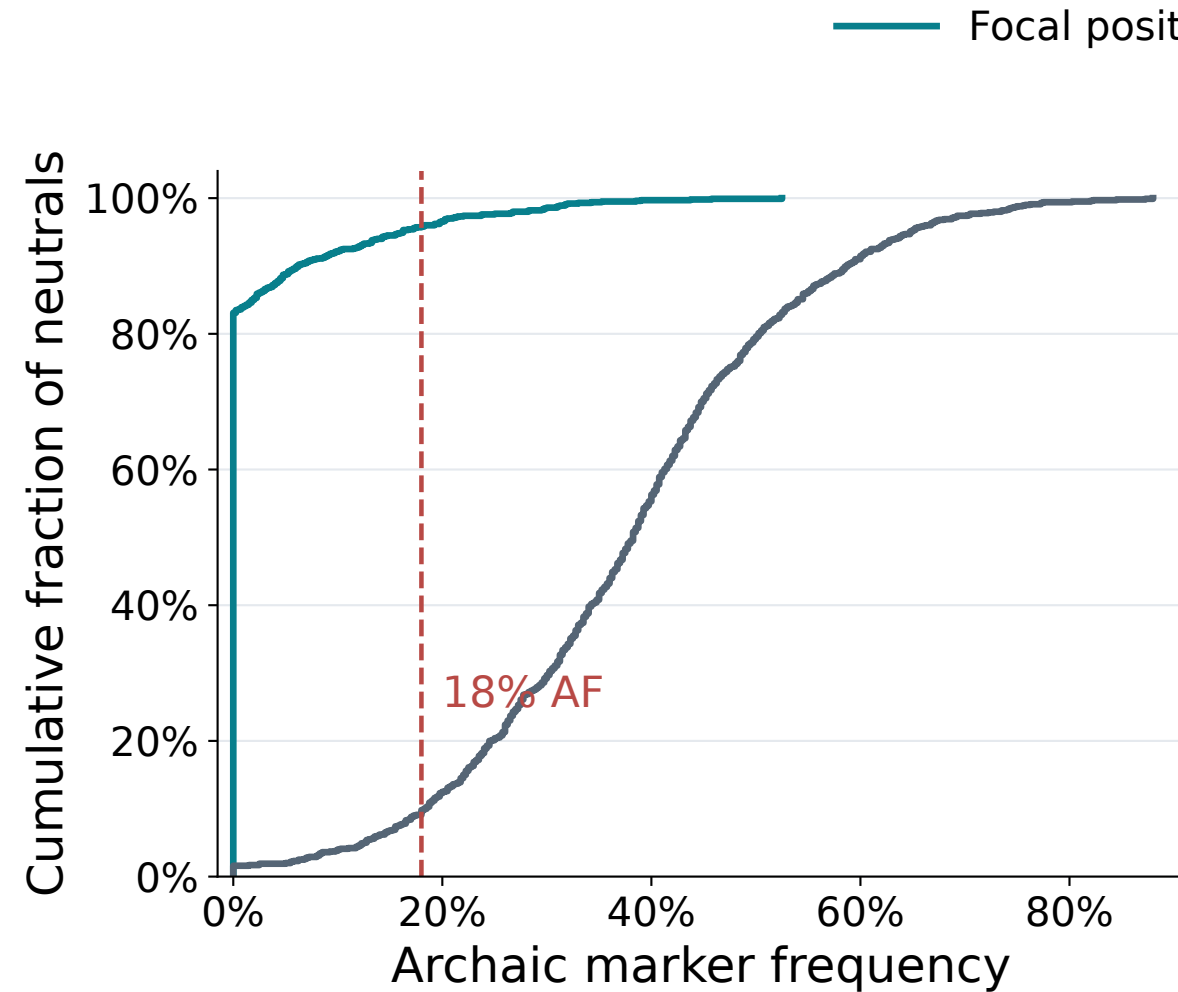
400 haplotypes; 10,000 fixed sampled pairs; isolated 10 Mb

$$S_T = \text{frac_recent}_T(\text{ALT/ALT}) \times \frac{n_{\text{ALT/ALT}}}{10,000}$$

Posterior-mean TMRCA hard calls | fixed-coordinate neutral calibration

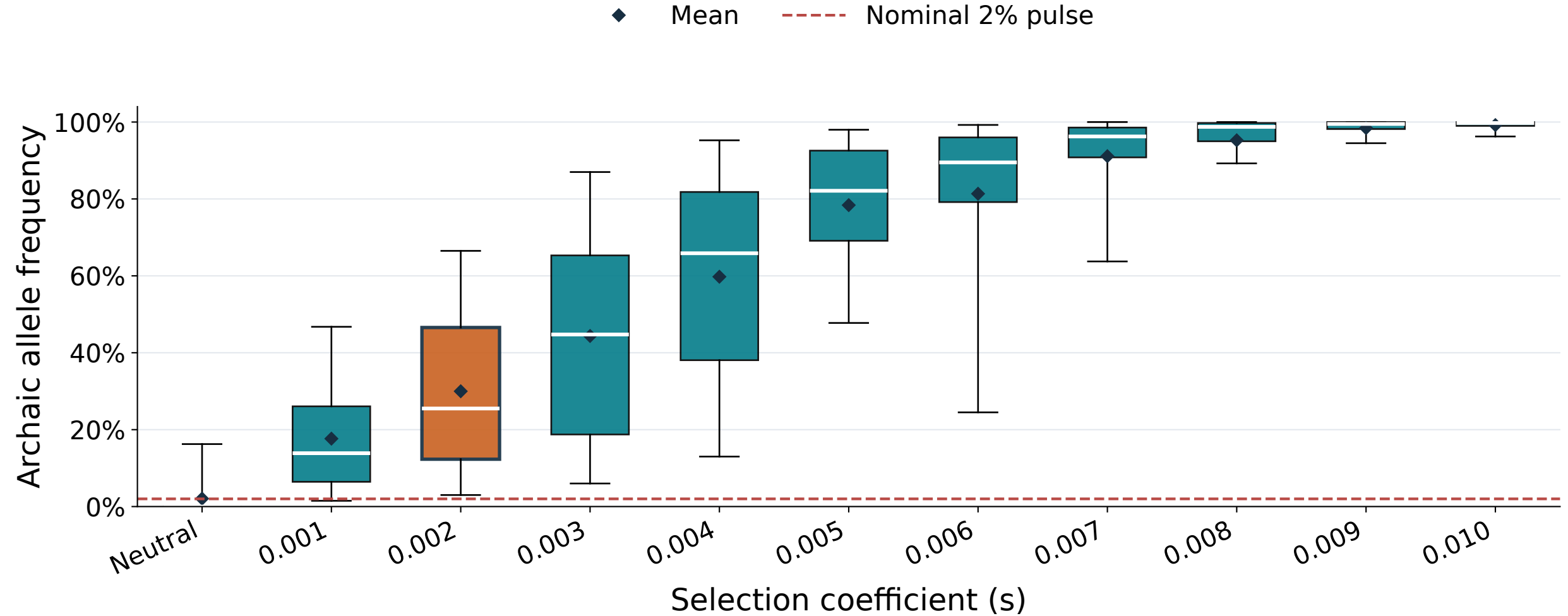
The null must match the position being tested

Same 1,000 neutral simulations; different spatial questions



Selection shifts local archaic-allele frequency

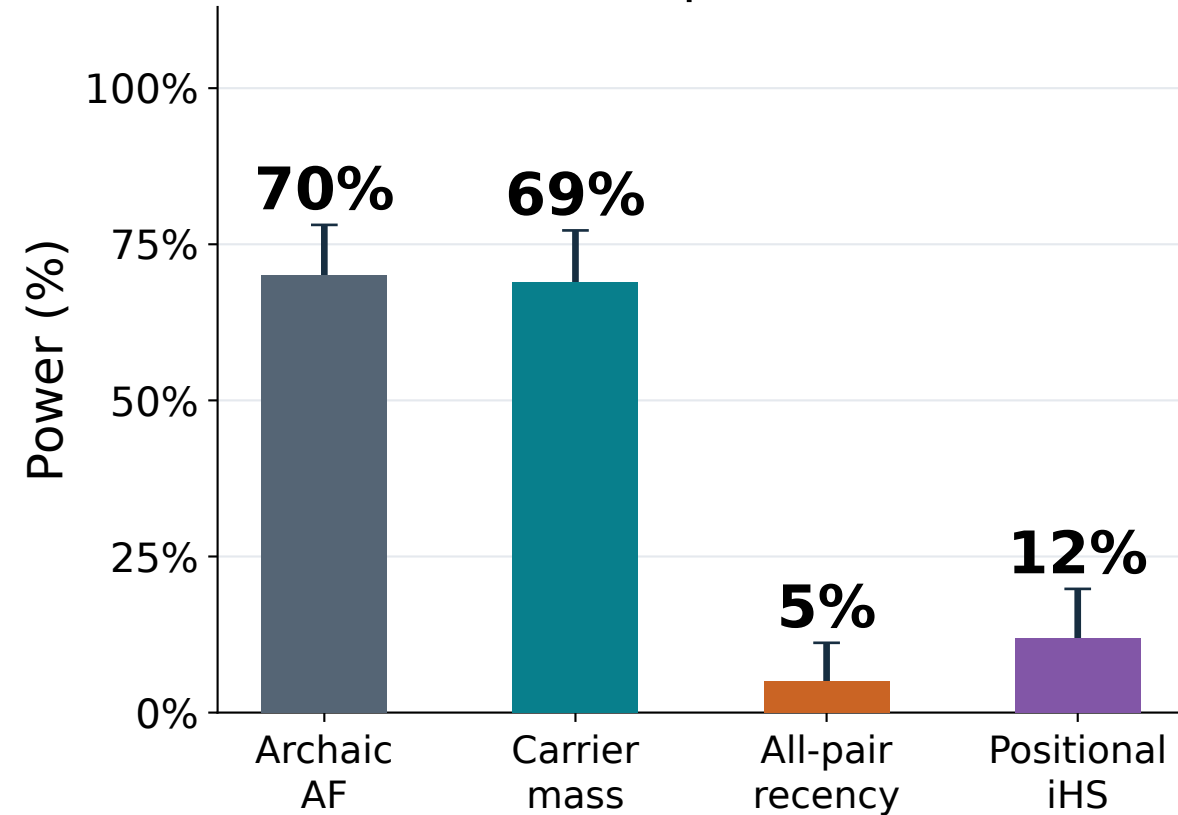
$s=0.002$ highlighted | genotype-based context across the completed selection grid



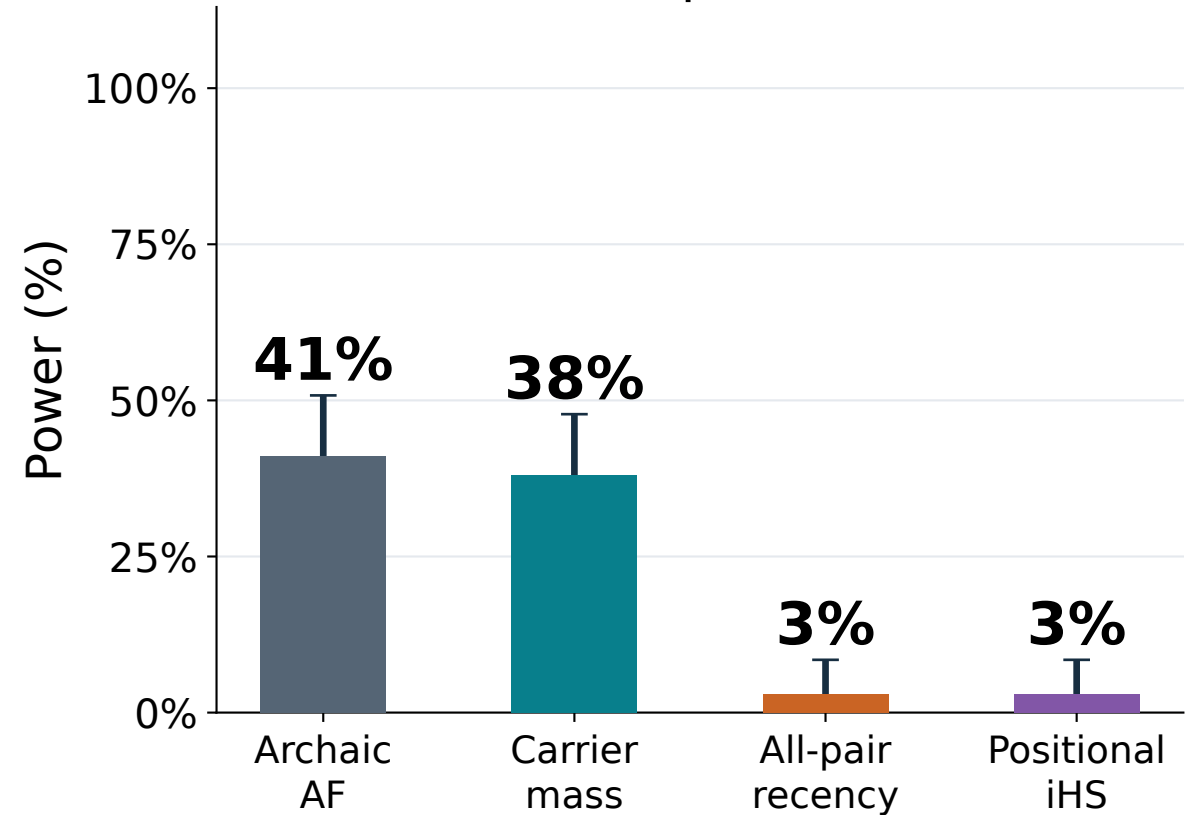
Gamma-SMC power at $s = 0.002$

T = 50 kya | 100 selected replicates | error bars: 95% Wilson intervals

Nominal $p \leq 0.05$

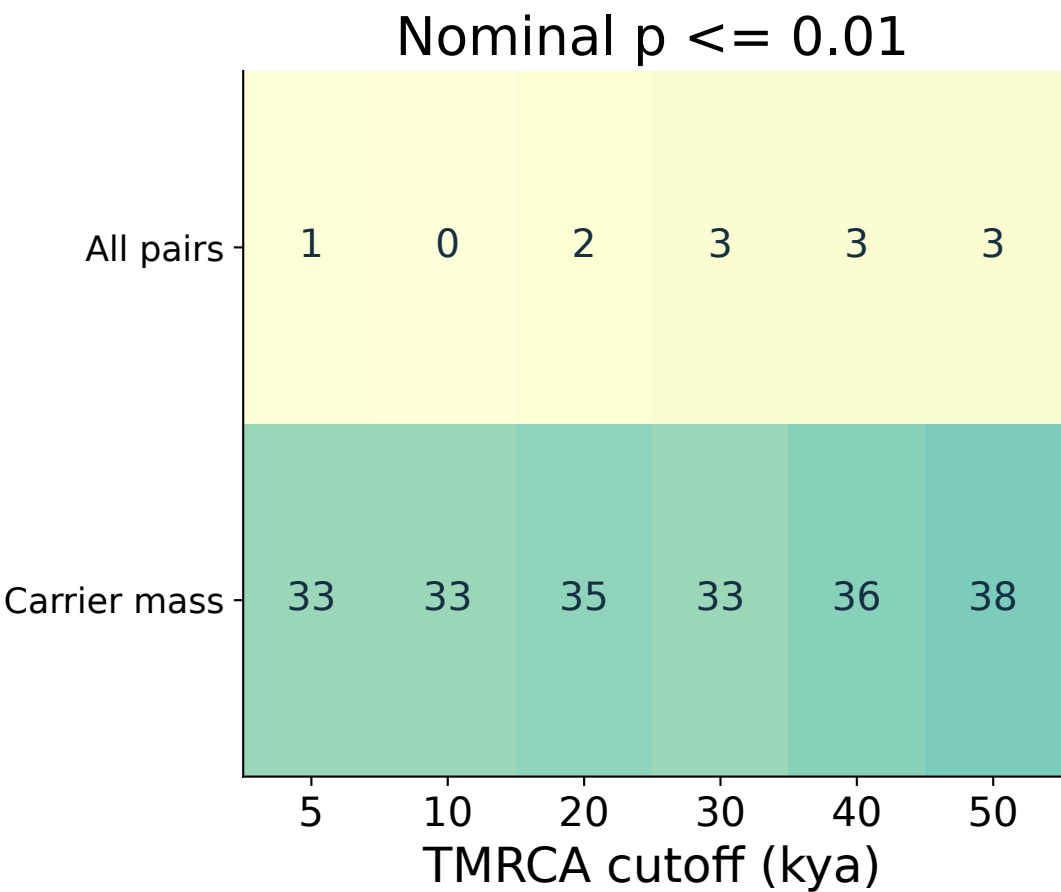
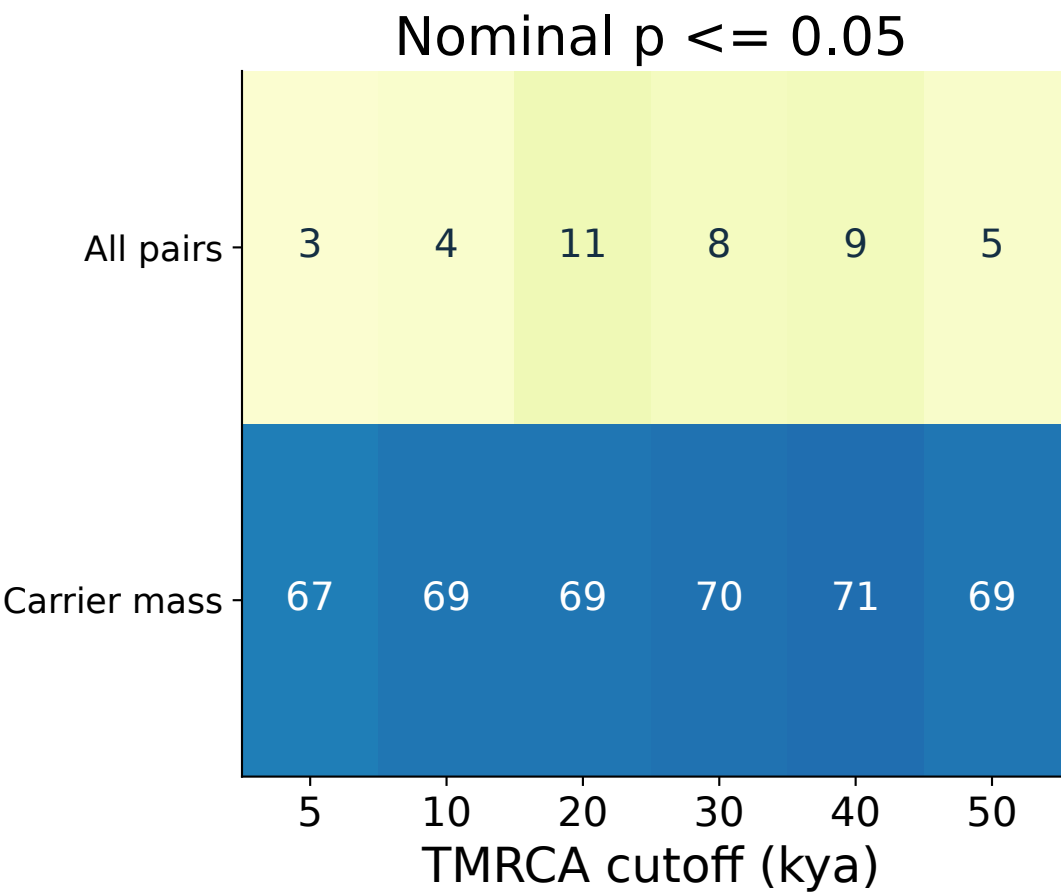


Nominal $p \leq 0.01$



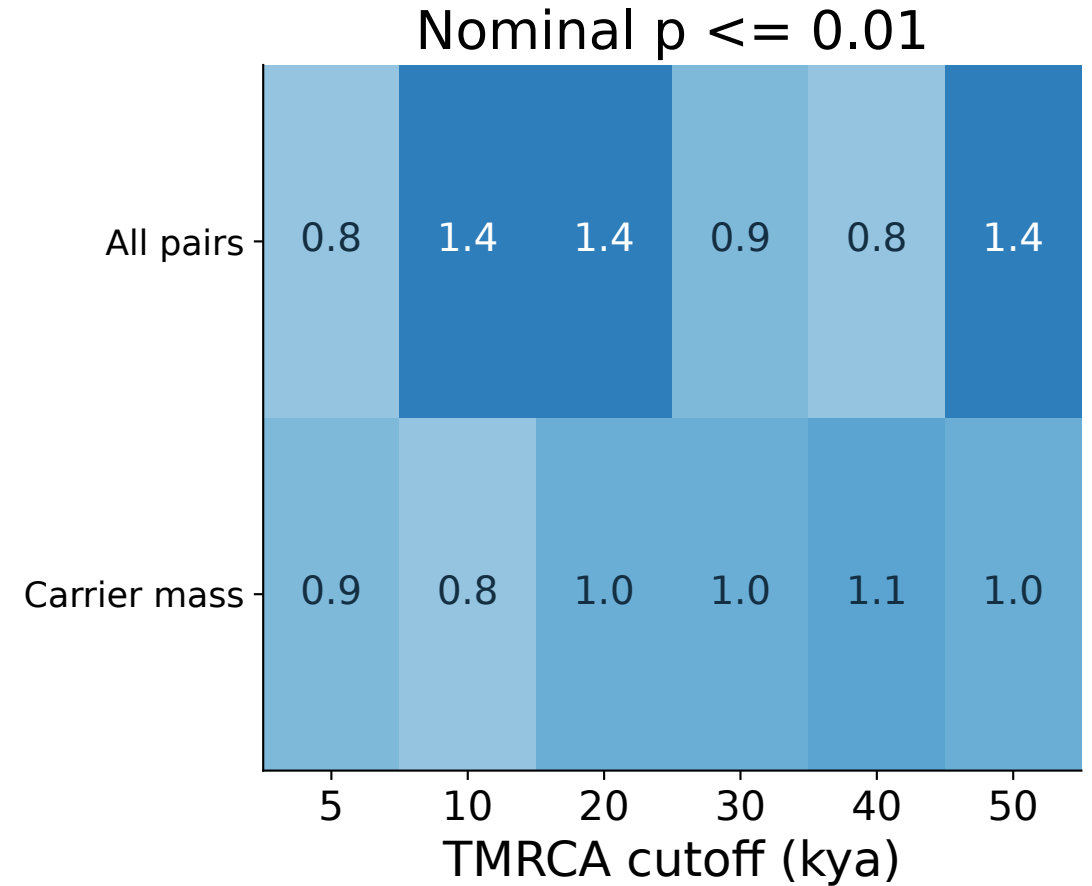
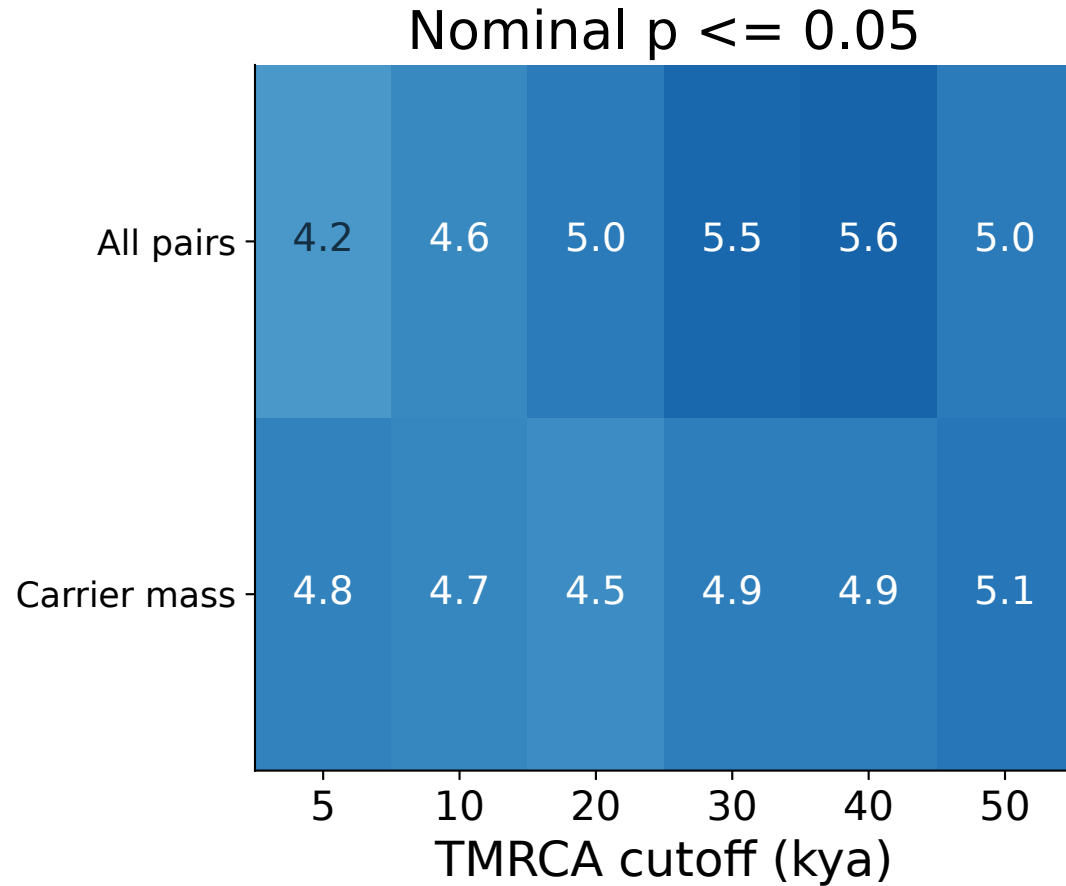
TMRCA cutoffs change decoded power

s = 0.002 | 100 selected replicates per cell | percentages



Decoded neutral false-positive rates

$s = 0.002$ | 1,000 neutral positions

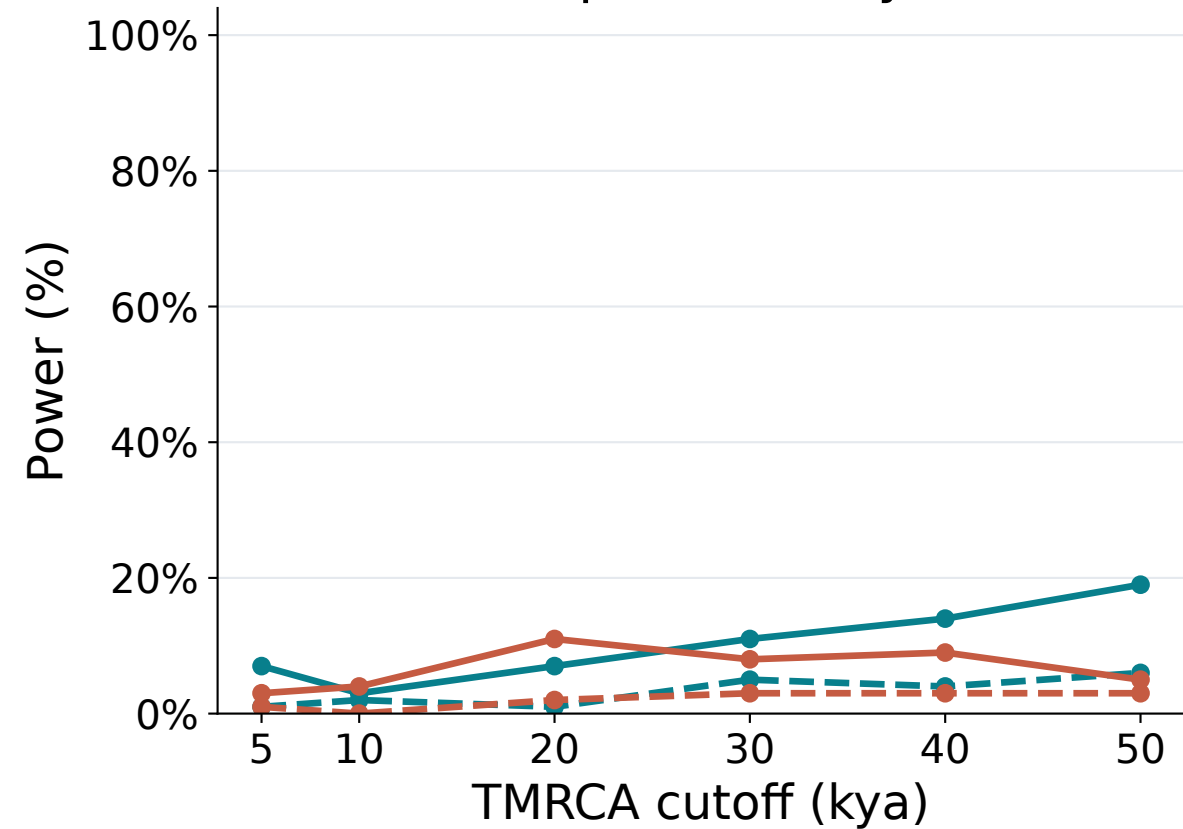


Validation: true versus decoded TMRCA

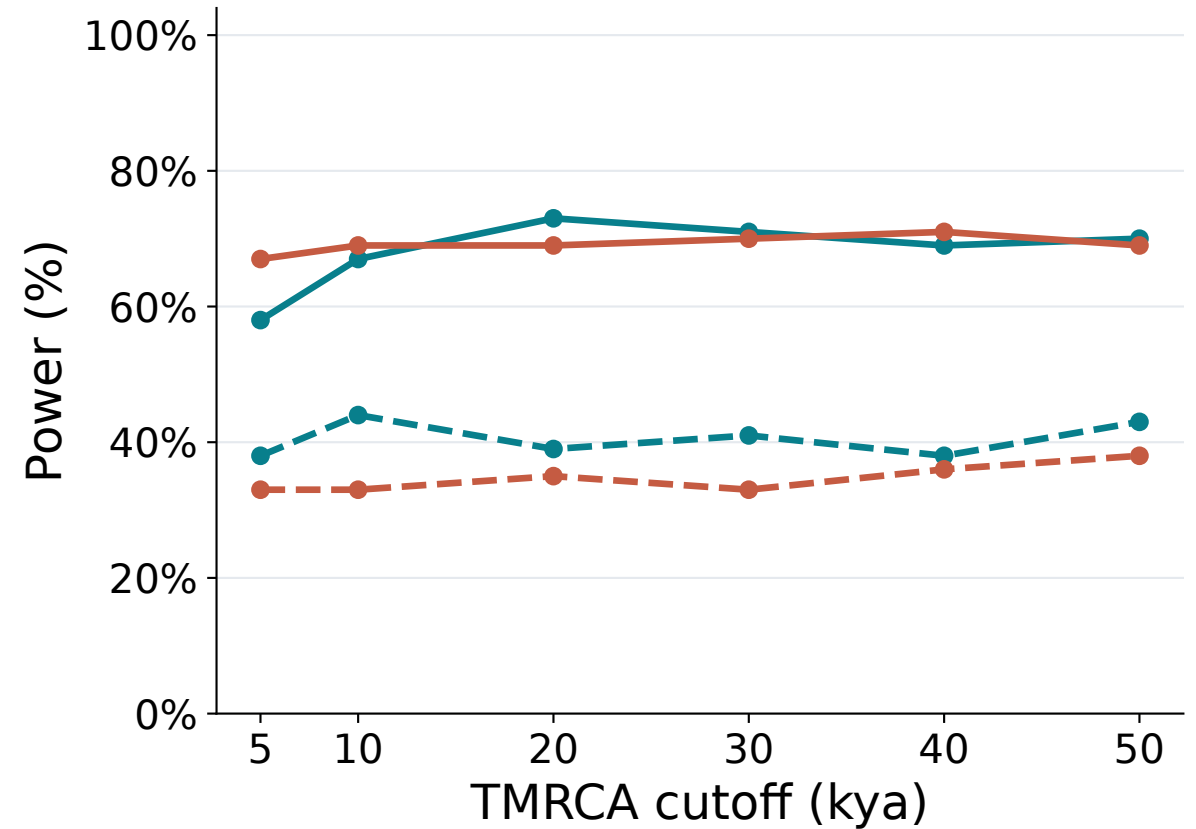
$s = 0.002$ | same saved genealogies, selected allele, and 10,000 pairs

—●— Truth, $p \leq 0.05$ -●- Truth, $p \leq 0.01$ —●— Decoded, $p \leq 0.05$ -●- Decoded, $p \leq 0.01$

All-pair recency



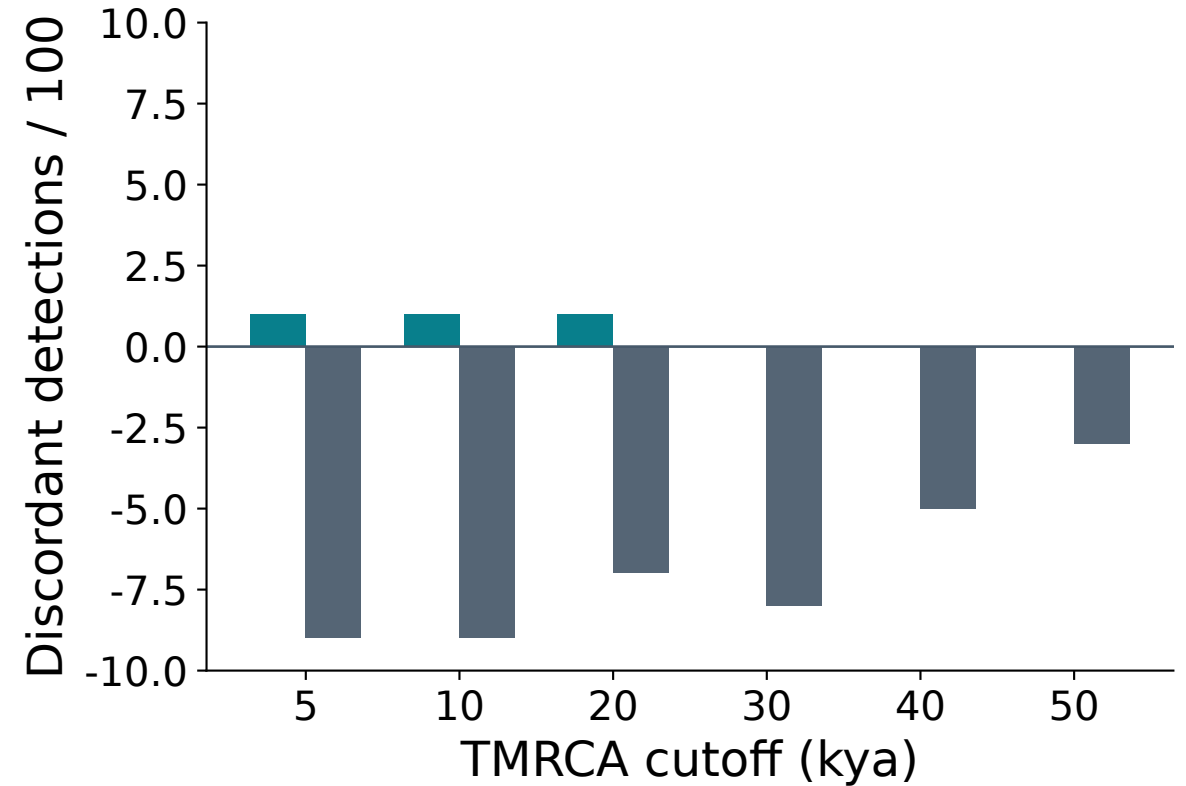
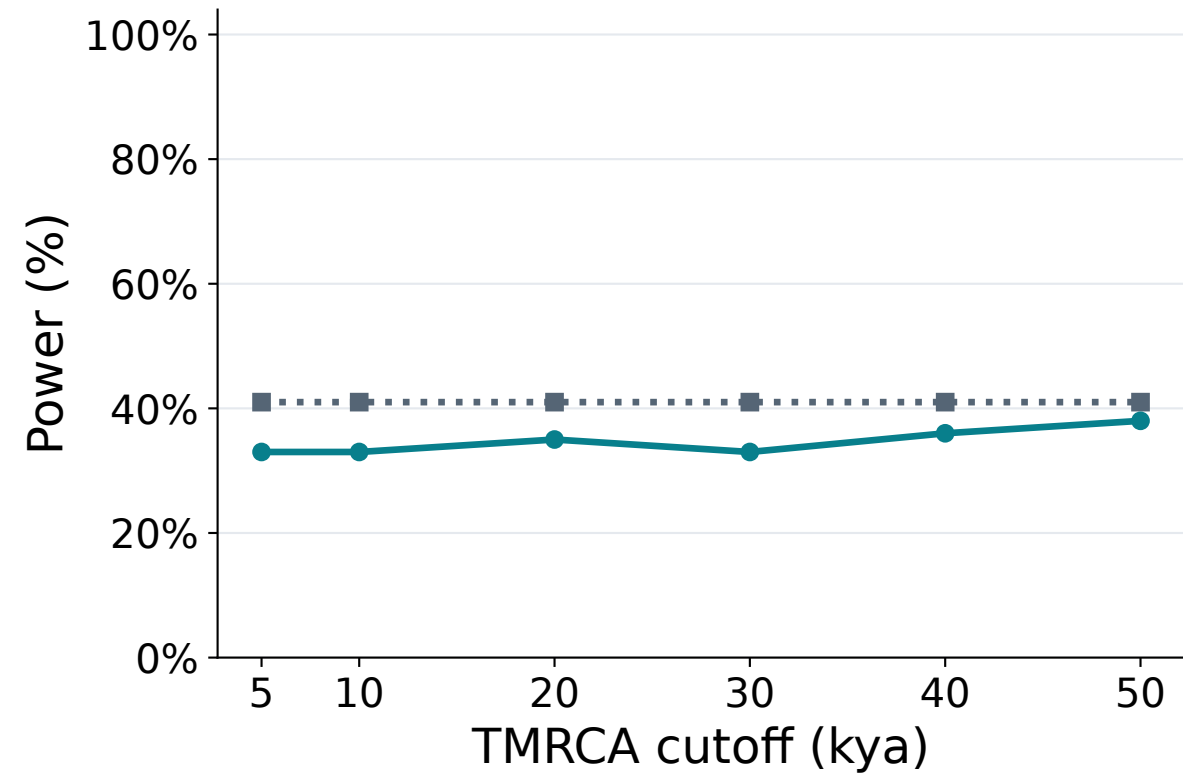
ALT/ALT carrier mass



Does decoded coalescence add to archaic AF?

$s = 0.002$ | nominal $p \leq 0.01$ | paired comparison of the same 100 replicates

··■·· Archaic AF ■ Carrier only (+)
—●— Decoded carrier mass ■ AF only (-)

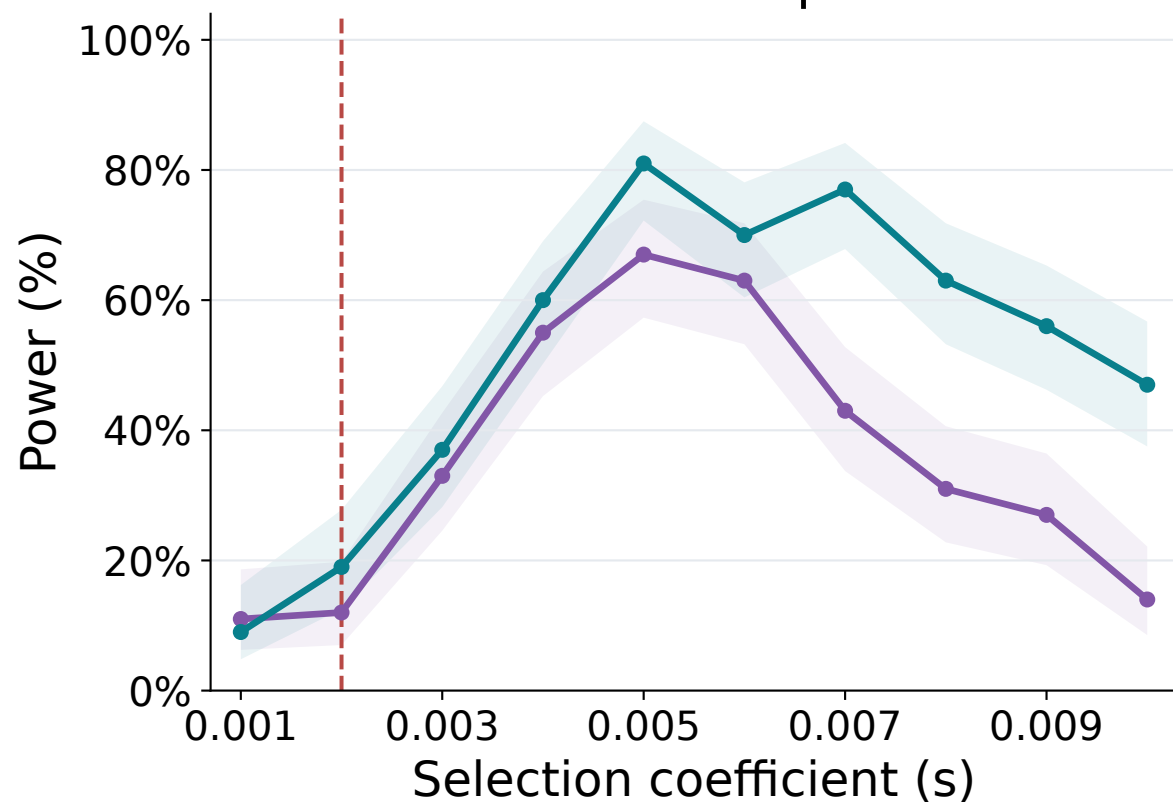


iHS is strongest before the target approaches fixation

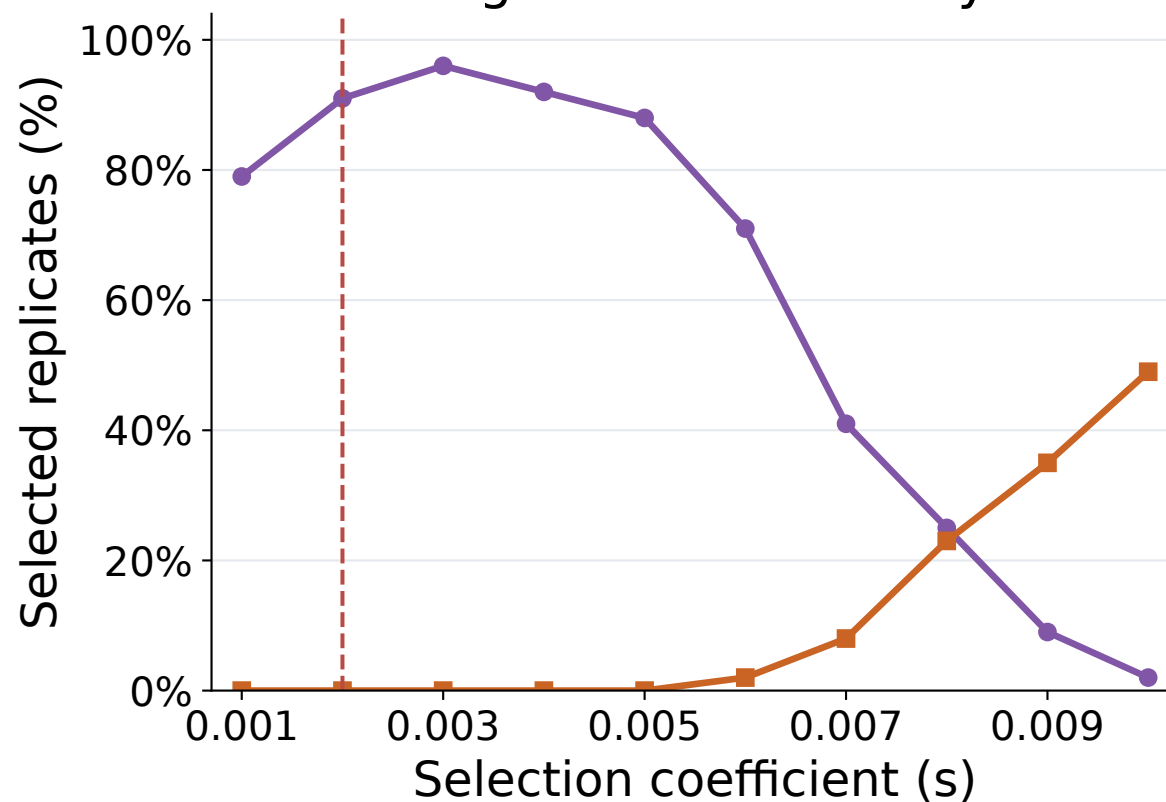
$s=0.002$ marked | genotype-based context; all selected replicates retained

- Positional iHS
- Fixed 100-kb window
- Focal SNP has finite iHS
- Selected allele fixed

Power at nominal $p \leq 0.05$



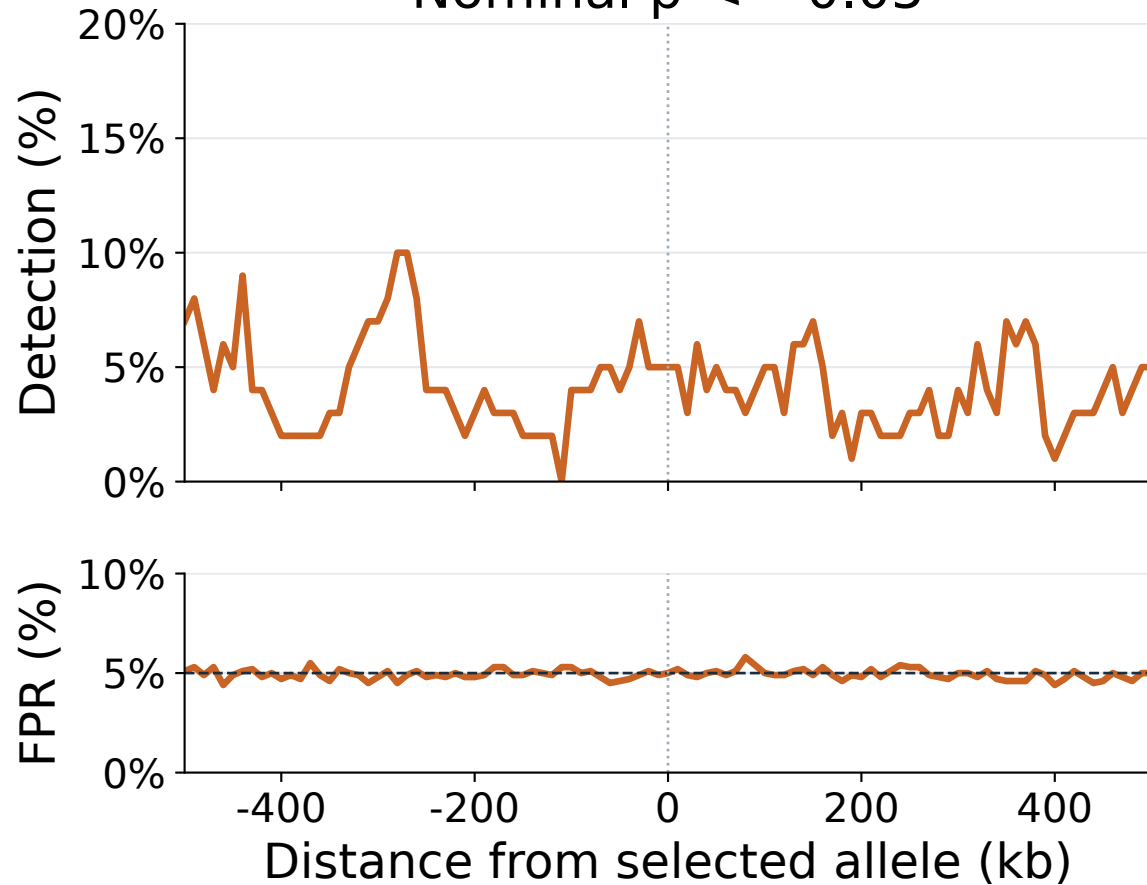
Target-site availability



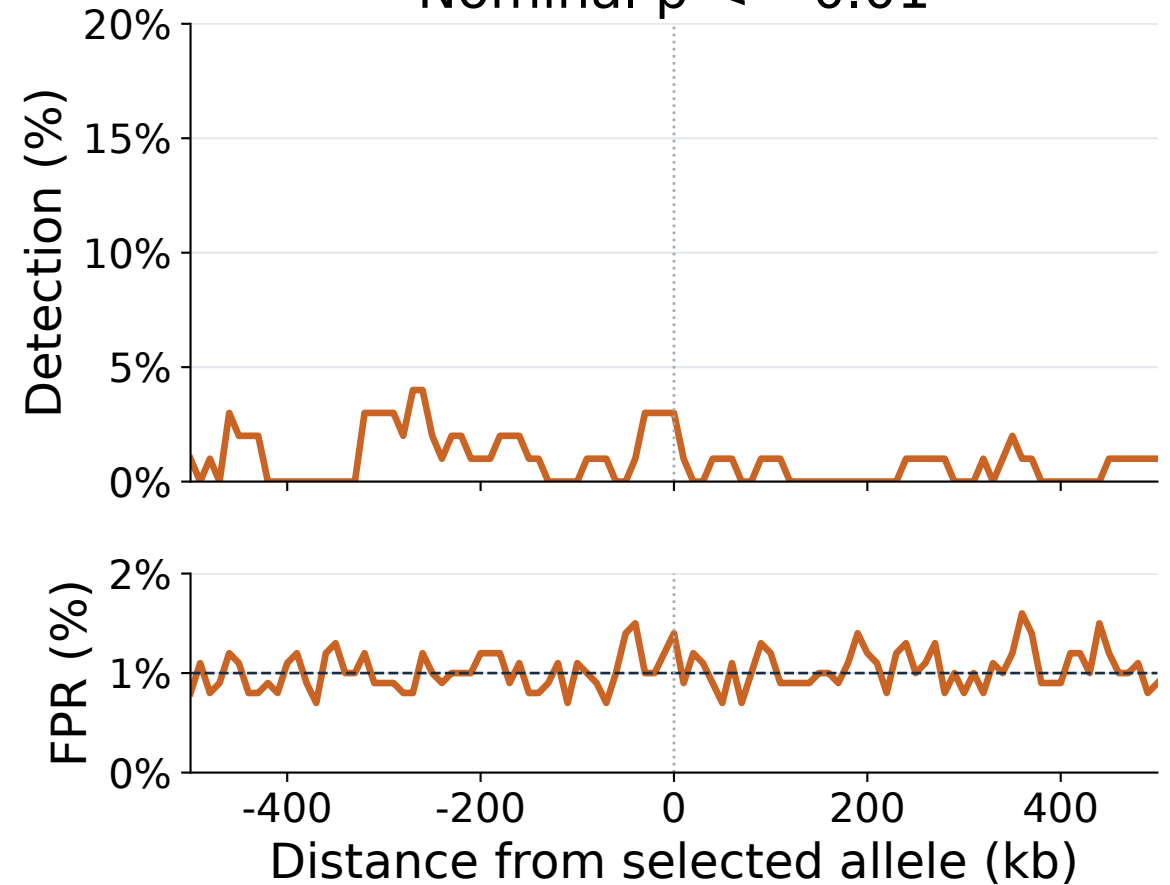
Spatial extent of the decoded all-pair signal

$s = 0.002$ | $T = 50$ kya | each stride uses its own matched neutral coordinate

Nominal $p \leq 0.05$

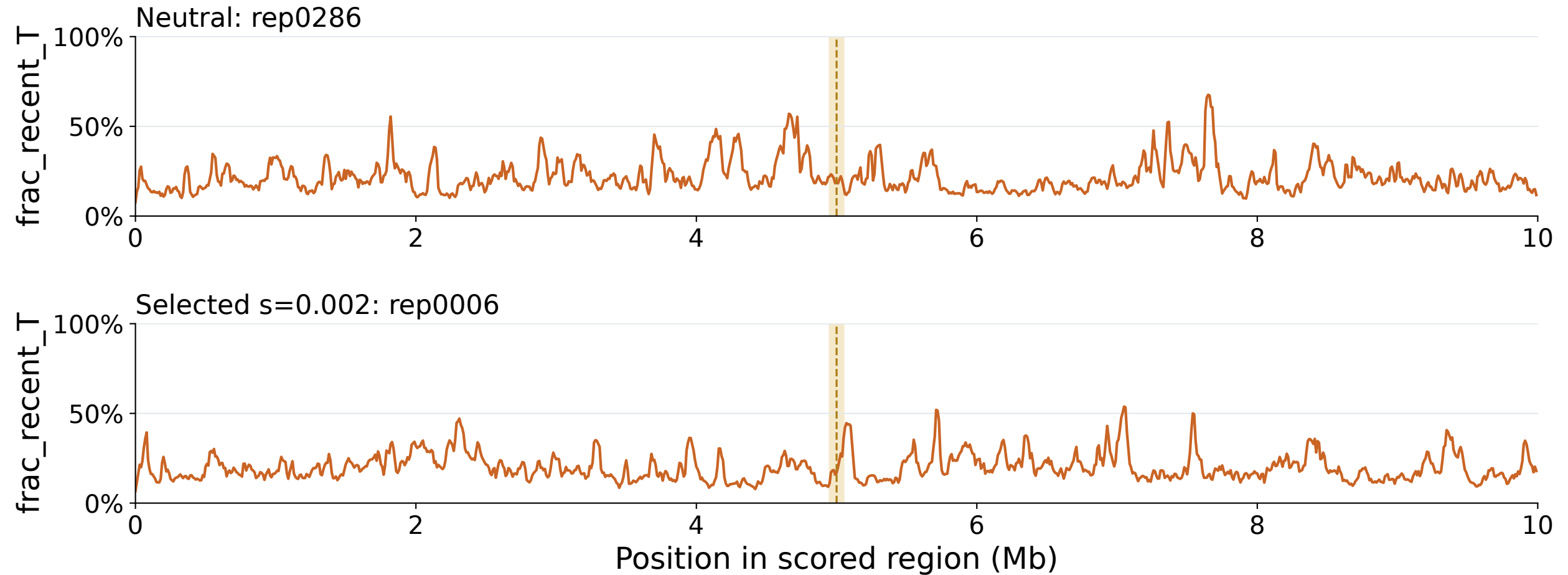


Nominal $p \leq 0.01$



Example decoded all-pair profiles

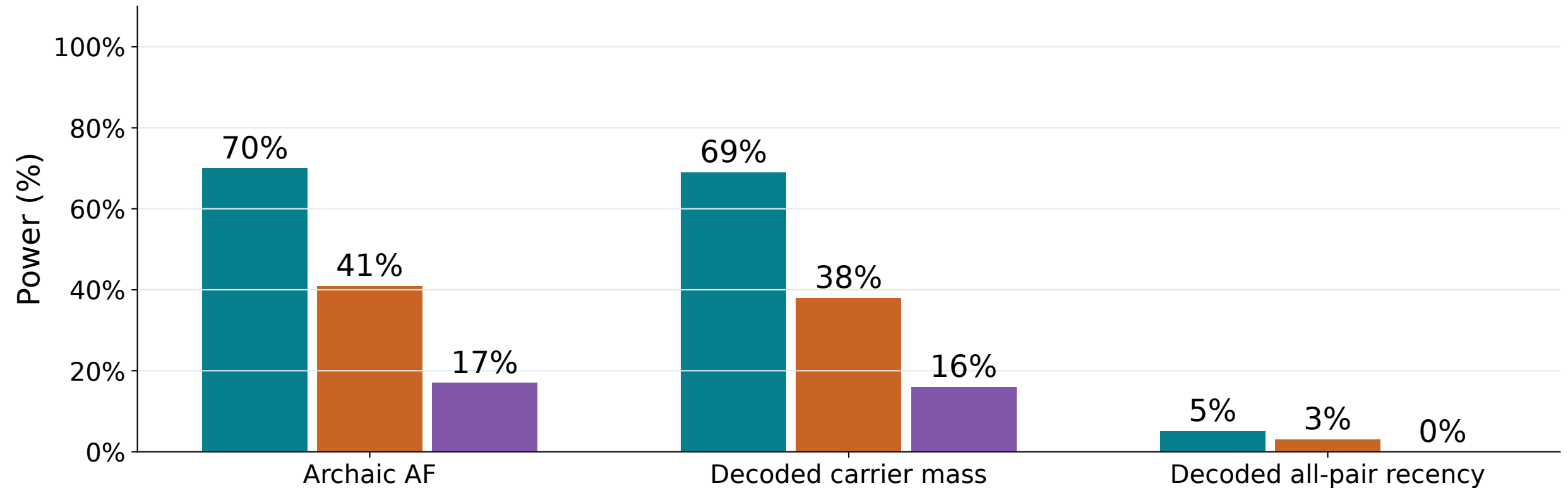
Examples chosen by median scores or allele frequency, independently of significance



Power at stricter positional thresholds

s = 0.002 | decoded TMRCA < 50 kya | p=0.001 uses pooled ranks

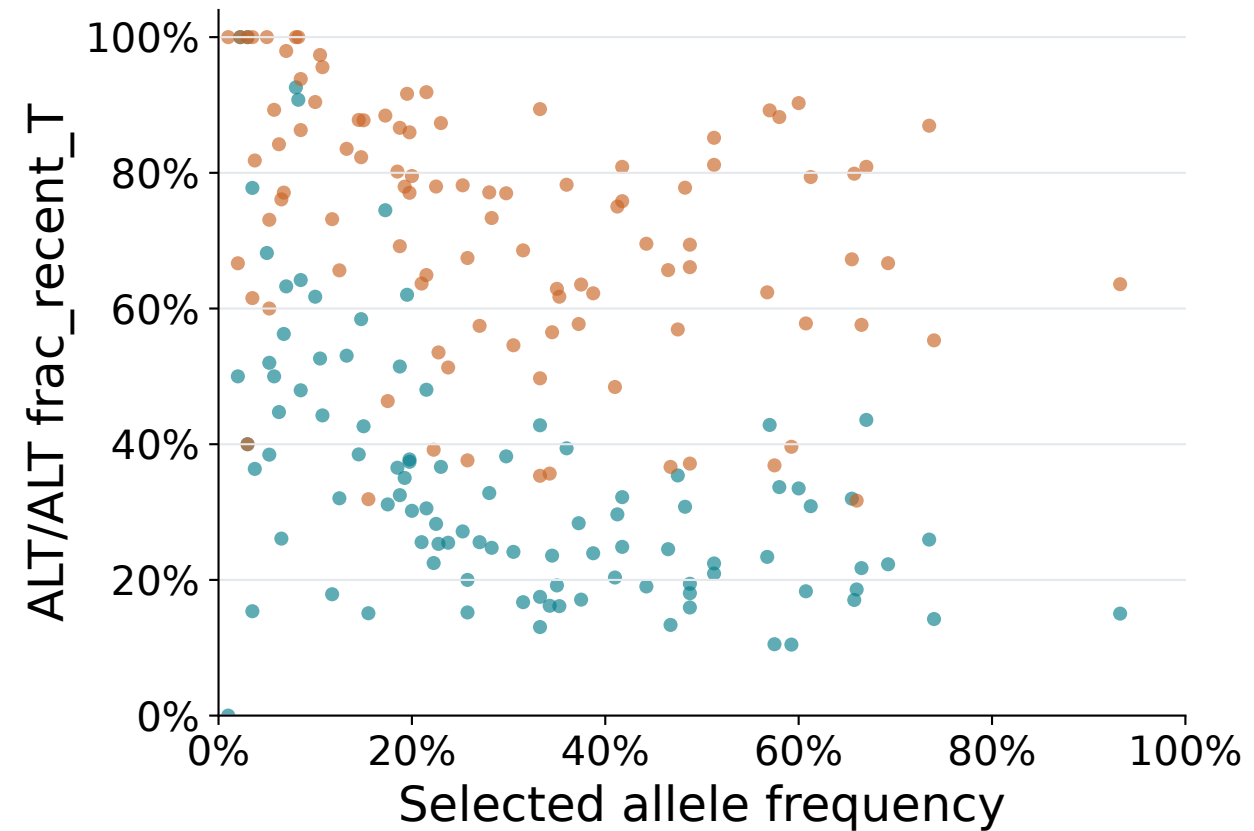
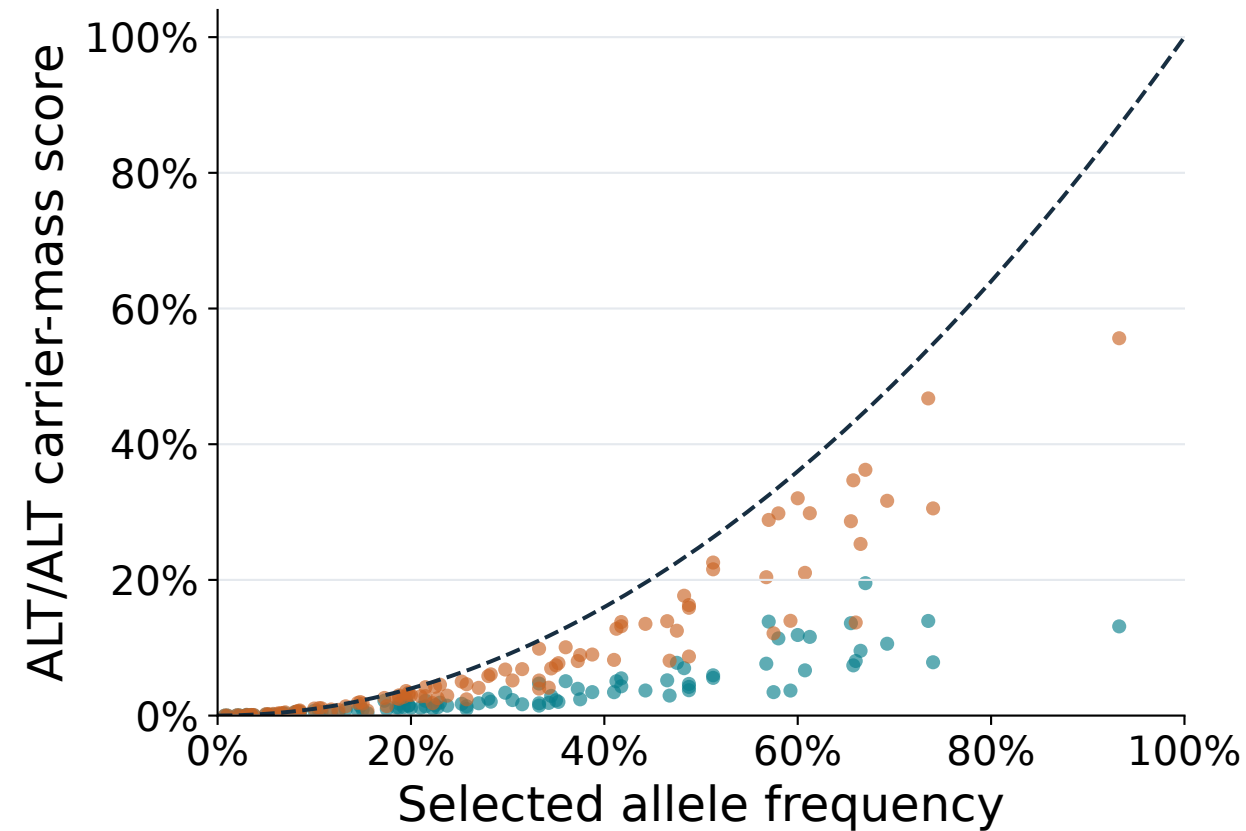
■ p ≤ 0.05 (5-fold) ■ p ≤ 0.01 (5-fold) ■ p ≤ 0.001 (pooled)



Focal carrier mass combines frequency and recency

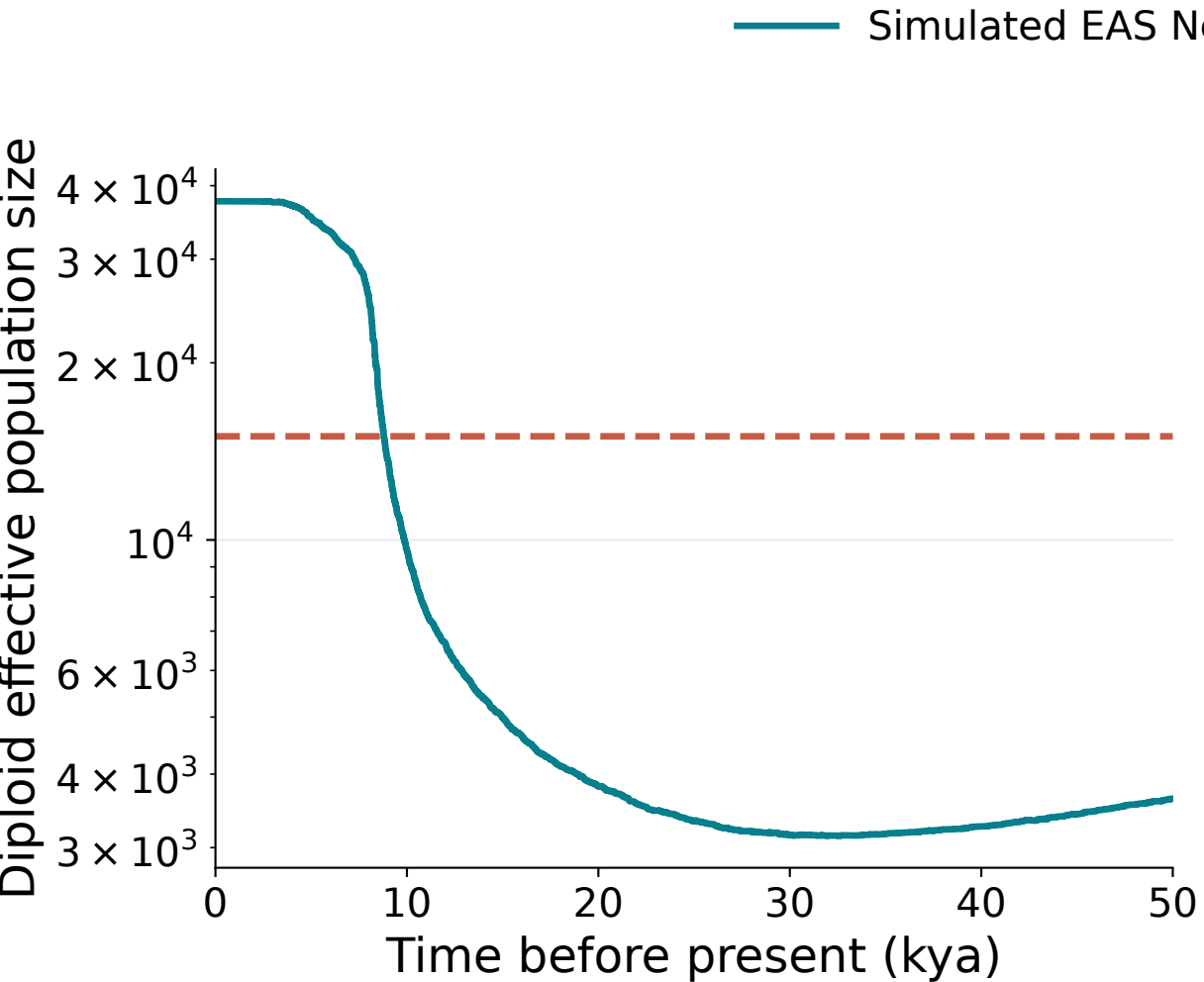
$s = 0.002$ | Gamma-SMC decoded | one dot per saved selected replicate

● T=20 kya ● T=50 kya ----- AF squared (approximation)



Demography is shared by neutral and selected simulations

The generative EAS history varies with time; the current decoder uses a fixed Ne



Archaic branch

Split: 700 kya

Baseline N_e : 3,600

Pulse bottleneck: $N_e=10$
for 100 generations

Mutation rate: 1.25×10^{-8}

Recombination rate: 1×10^{-8}

Generation time: 25 years

Gamma-SMC recency among archaic ALT/ALT pairs

Decoded focal TMRCA | pairs pooled within each class | $s=0.002$ selected versus neutral

